

1200 V Fully-Vertical GaN-on-Si Power MOSFETs

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Abstract—We report 1200 V fully-vertical GaN-on-Si trench MOSFETs with fluorine implanted termination (FIT-MOS). The FIT region with negative fixed charges becomes resistive and naturally isolates the discrete devices, replacing the conventional mesa etching termination (MET), eliminating the electric field crowding effect at the mesa edges, therefore boosting the breakdown voltage of the FIT-MOS to 1277 V from 567 V of the MET-MOS. Moreover, the as-fabricated FIT-MOS exhibits a threshold voltage (V_{TH}) of 3.3 V, an ON/OFF ratio of $\sim 10^7$, together with a specific ON-resistance ($R_{ON,SP}$) of $5.6 \text{ m}\Omega\cdot\text{cm}^2$. These results show great potential of cost-effective GaN-on-Si vertical transistors for kV-class applications.

Index Terms—GaN-on-Si, vertical transistors, power MOSFETs, fully-vertical, breakdown voltage, fluorine implanted termination, electric field.

I. INTRODUCTION

WIDE-BANDGAP semiconductors such as GaN and SiC possess immense potential for applications in high-efficiency and high-density power systems. While GaN-based transistors have been extensively deployed for power applications rated at 100-650 V [1], SiC-based transistors have predominantly found their commercial footing in high-voltage domains exceeding 1200 V [2]. Nevertheless, it remains uncertain which of these two counterparts would emerge as the optimal choice for the market applications within the 650-1200 V range, which is characterized by intense competition in both cost and performance aspects [3]. Although significant progress has been made in the development of SiC based transistors, the relatively high cost of SiC

substrates limits their competitiveness in this voltage range from the economic vantage point. Fortunately, the advancements in GaN hetero-epitaxy technology on low-cost and large-scale silicon (Si) substrates point to tremendous promise for fabricating cost-effective and high-performance GaN transistors [4]. Benefiting from the high-density two-dimensional electron gas (2DEG) induced at the AlGaN/GaN heterointerface, GaN-based lateral high-electron-mobility-transistors (HEMTs) are highly advantageous for high-frequency and low-voltage power applications. However, extending the breakdown voltage of GaN-on-Si lateral HEMTs beyond 650 V remains an obstacle for the lateral architectures due to their inherent scalability limitations [5]. In contrast, thanks to the design flexibility of the vertical topologies, vertical GaN power transistors present significant advantages in kV-class blocking capability by increasing the thickness of the drift layer without enlarging the footprint of the devices. In recent years, several 1200 V vertical P-i-N diodes have been successfully demonstrated with excellent blocking capability [6], [7] and avalanche capability [8], validating the feasibility of vertical GaN-on-Si topology for high-voltage and high-power applications.

Liu et al. [9] reported the first quasi-vertical GaN-on-Si trench MOSFETs with a $4 \mu\text{m}$ thick GaN drift layer, exhibiting a hard breakdown voltage of 645 V. Subsequent investigations focused on optimizing the p-GaN channel region and gate dielectric layers to improve the ON-state performance and reliability of the quasi-vertical GaN-on-Si trench MOSFETs [10], [11], [12]. To eliminate the current crowding effect in the devices with quasi-vertical configuration, Khadar et al. [13] demonstrated the first fully-vertical GaN-on-Si trench MOSFETs, whose drain pads were deposited on the backside n^+ -GaN layer exposed by selective removal of the buffer and Si substrate. Biswas et al. [14] reported fully-vertical GaN-on-Si transistors with a conductive buffer layer, which eliminates the complicated substrate engineering process and simplifies the fabrication process. However, due to the absence of effective termination structures and high-quality drift layers, the breakdown voltage (BV) of the reported vertical GaN-on-Si MOSFETs remains below 650 V, which is insufficient for kV-class applications in power systems.

In this letter, we demonstrate 1200 V fully-vertical GaN-on-Si trench MOSFETs by incorporating fluorine implanted termination (FIT). The traditional mesa etching termination (MET) is replaced by the resistive FIT structure, which effectively isolates the discrete devices and alleviates

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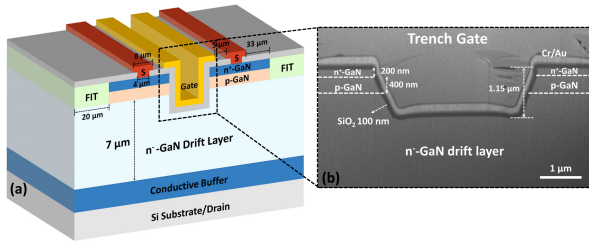


Fig. 1. (a) Schematic structure and (b) cross-sectional SEM image (trench gate region) of the fully-vertical GaN-on-Si trench MOSFETs with fluorine implanted termination (FIT-MOS).

the crowded electric field near the termination region, therefore resulting in an enhanced BV to 1277 V (FIT-MOS) from 567 V (MET-MOS). In addition, the FIT-MOS exhibits an enhancement operation mode with a threshold voltage (V_{TH}) of 3.3 V, an ON/OFF ratio of $\sim 10^7$, together with a low specific ON-resistance ($R_{ON,SP}$) of 5.6 mΩ·cm², as well as a high ON-state current density of 8 kA/cm². These results pave the way for the development of kV-class power electronic systems based on the cost-effective vertical GaN-on-Si transistors.

II. DEVICE ARCHITECTURE AND FABRICATION

Fig.1(a)-(b) depict the schematic structure of the fully-vertical GaN-on-Si FIT-MOS and cross-sectional scanning electron microscope (SEM) image of the trench gate region. With implanted highly electronegative fluorine (F) ions, the originally conductive n⁺-GaN/p-GaN layers in the FIT region become resistive, which effectively isolates the discrete FIT-MOS devices and eliminates the need of traditional METs [15]. The depth of the gate trench in the FIT-MOS is 1.15 μm, reaching the n⁻-GaN drift layer. A conformal gate stack can be observed, consisting of a 100 nm thick SiO₂ gate dielectric layer and 50/150 nm thick Cr/Au gate metal layers.

The fabrication process of the FIT-MOS began with a gate trench etching process using SiO₂ as hard mask, followed by 850°C rapid thermal annealing (RTA) in N₂ ambient for 20 minutes to activate the acceptors in the buried p-GaN layer. The dry etching damage was repaired through wet treatment in 25% Tetra Methyl Ammonium Hydroxide (TMAH) solution at 85°C for 3 h. Triple-energy F ion-implantation (energy level/dose: 240 keV/4 × 10¹⁴ cm⁻², 140 keV/2 × 10¹⁴ cm⁻², 80 keV/1.2 × 10¹⁴ cm⁻²) was conducted by ion implanter (NV-GSD-HE from Sumitomo Eaton Nova) masked with thick photoresist (AZ4620: ~ 6.5 μm) to form the FIT region. Subsequently, a 100 nm thick SiO₂ gate dielectric layer was deposited using atomic layer deposition (ALD), followed by source contact hole opening via reactive ion etching (RIE). Finally, a bilayer Cr/Au stack was deposited to form the gate and source electrodes [9]. The low-resistance Si substrates served as the drain electrodes. The MET-MOS shares similar process flow with the FIT-MOS, except that the MET-MOS utilizes mesa etching process to isolate discrete devices.

III. RESULTS AND DISCUSSION

Fig. 2(a)-(b) present the cross-sectional SEM image and the secondary ion mass spectrometry (SIMS) profiles of the N-P-N epitaxial structure for the fully-vertical trench MOSFETs. From top to bottom, the N-P-N structure

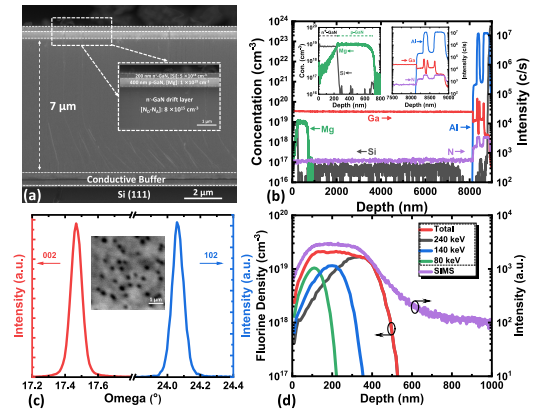


Fig. 2. (a) The cross-sectional SEM image and (b) the SIMS profiles of N-P-N epitaxial structures for the fully-vertical trench MOSFETs. (c) Omega rocking curves for (002) orientation and (102) orientation. The inset shows the CL image of the as-grown GaN layer on Si. (d) The distribution of F ions from TRIM simulation and SIMS measurement.

consists of a 20 nm thick n⁺⁺-GaN layer (Si: $\sim 1 \times 10^{19}$ cm⁻³), a 200 nm thick n⁺-GaN source contact layer (Si: $\sim 5 \times 10^{18}$ cm⁻³), a 400 nm thick p-GaN channel layer (Mg: $\sim 1 \times 10^{19}$ cm⁻³), and a 7 μm thick n⁻-GaN drift layer (N_D-N_A: $\sim 8 \times 10^{15}$ cm⁻³) grown on 6-inch Si substrates with a conductive buffer layer by metal-organic chemical vapor deposition (MOCVD). The conductive buffer layer, composed of AlGaIn/AlN multilayers, enables a fully vertical current path while eliminating the need for complex substrate engineering process to achieve the fully-vertical configuration [13], [14], [16]. On the other hand, the AlGaIn/AlN based buffer layer provides sufficient compressive stress upon the GaN drift layer above, which effectively compensates the tensile stress and avoids possible cracks during the cooling down process from high growth temperature [4], [14]. The estimated threading dislocation density (TDD) from the full width at half maximum (FWHM) of the X-ray omega rocking curves (Fig. 2(c)) is 3.0×10^8 cm⁻² through the following equation [17]:

$$D = \frac{\beta_{(002)}^2}{9b_1^2} + \frac{\beta_{(102)}^2}{9b_2^2} \quad (1)$$

where β is the FWHM and $\vec{b}$ is the Burgers vector. A similar TDD of 1.4×10^8 cm⁻² is characterized by cathodoluminescence (CL) measurement, as shown in the inset of Fig. 2(c). Fig. 2(d) shows the profiles of the F ions obtained from Transport of Ion in Matter (TRIM) simulation and SIMS measurement, displaying a uniform F ion distribution in the FIT structure.

Fig. 3(a) illustrates the transfer I_D - V_G characteristics of the FIT-MOS in linear and log scales at $V_{DS} = 10$ V. The FIT-MOS exhibits enhancement operation mode operation with a positive V_{TH} of 3.3 V (extracted at $I_{DS} = 1$ A/cm²), an ON/OFF ratio of $\sim 10^7$, as well as a hysteresis of ~ 0.5 V [13]. Fig. 3(b) exhibits the output I_D - V_D characteristics of the fabricated FIT-MOS, revealing a high ON-state current density of 8 kA/cm². The FIT-MOS exhibits a relatively low $R_{ON,SP}$ of 5.6 mΩ·cm² calculated from the linear region.

Fig. 4 compares the OFF-state breakdown characteristics of the FIT-MOS and MET-MOS. The gate leakage (I_G) remains

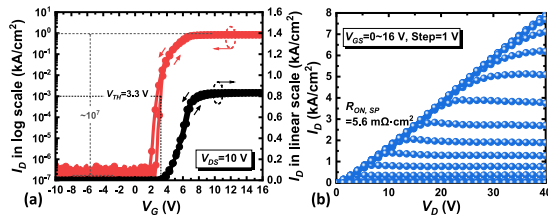


Fig. 3. (a) The transfer and (b) output I-V characteristics of the fabricated fully-vertical GaN-on-Si FIT-MOS. I_D is normalized by the trench area [9], [18], [19].

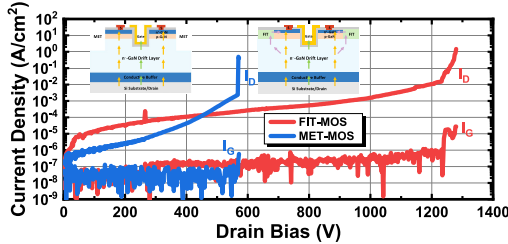


Fig. 4. OFF-state I-V characteristics at $V_{GS} = 0$ V of the fabricated fully-vertical GaN-on-Si FIT-MOS and MET-MOS. The inset shows the potential leakage paths of the FIT-MOS and MET-MOS.

below 10^{-6} A/cm² before breakdown, indicating excellent stability of the gate dielectric stacks at high drain bias. The FIT-MOS exhibits a high breakdown voltage of 1277 V, while premature breakdown occurs in the MET-MOS at 567 V. The FIT-MOS exhibits a larger OFF-state current density at low V_{DS} than that of the MET-MOS due to the additional vertical leakage paths associated with the FIT structure as shown in the inset of Fig. 4 [20], which can be verified by the pad-to-pad leakage current density measured from the test structures with FIT and MET isolations, as shown in Fig. 5(a). Moreover, due to the existence of Ga vacancies in the FIT region after implantation as well as insufficient atomic bonding and energy confinement, the F ions may possibly diffuse in the GaN lattice and escape from the surface, which disturbs the thermal stability of F implanted devices [21]. Adopting optimized post-implantation annealing process can efficiently reduce the OFF-state leakage current density as well as enhance the thermal reliability of the FIT-MOS [15], [21].

To analyze the breakdown mechanism in the MET-MOS and FIT-MOS, TCAD simulation was carried out to calculate the two-dimensional electric field distribution as depicted in Fig. 5(b)-(c). The key physical models used in the simulation include the carrier drift-diffusion model, generation-recombination model, continuity and Poisson equations, and impact ionization model [22]. Crowded electric field can be observed from the MET structure at $V_{DS} = 400$ V in Fig. 5(b), with a peak electric field of 2.7 MV/cm at the mesa corner, degrading the breakdown voltage of the MET-MOS. As shown in Fig. 5(c), the crowded electric field near the edge termination can be effectively suppressed by the incorporated FIT structure. However, a crowded electric field is observed near the trench gate region, which can be effectively suppressed by gate shielding structure [22].

Fig. 6(a)-(b) benchmark the $R_{ON,SP}$ vs. BV as well as drift layer thickness (T_{drift}) vs. BV of our fabricated GaN-on-Si FIT-MOS with reported vertical GaN trench MOSFETs grown on Si [9], [10], [13], [23], [24], Sapphire (Sap) [25],

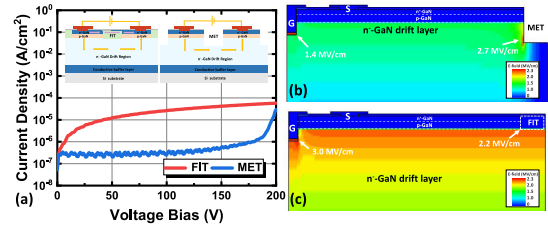


Fig. 5. (a) Pad-to-pad leakage current density from the test structures (inset) with FIT and MET isolation. Simulated electric field distribution in the (b) MET-MOS at $V_{DS} = 400$ V and (c) FIT-MOS at $V_{DS} = 1200$ V, respectively.

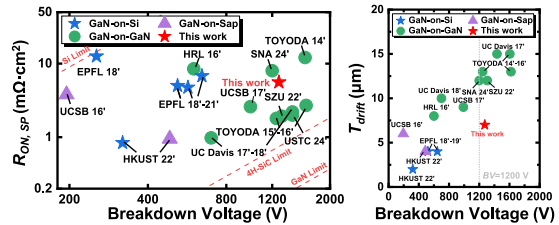


Fig. 6. Benchmark of (a) $R_{ON,SP}$ vs. BV and (b) drift-layer-thickness (T_{drift}) vs. BV for the fabricated GaN-on-Si fully-vertical FIT-MOS against with reported GaN vertical trench MOSFETs on Si [9], [10], [13], [23], [24], Sapphire (Sap) [25], [26] and GaN substrates [19], [27], [28], [29], [30], [31], [32], [33], [34], [35].

[26] and GaN [19], [27], [28], [29], [30], [31], [32], [33], [34], [35] substrates. The fabricated FIT-MOS exhibits state-of-the-art BV of 1277 V, a low $R_{ON,SP}$ of 5.6 mΩ·cm², and a high Baliga figure-of-merit ($BFOM = BV^2/R_{ON,SP}$) of 291 MW/cm², on foreign substrates (Si and Sap substrates), which is also comparable to those fabricated on native GaN substrates. Moreover, thanks to the high-efficiency fluorine implantation termination, the fully-vertical GaN-on-Si FIT-MOS with 7 μm thick drift layer demonstrates comparable BV to those 1200 V-class vertical GaN-on-GaN trench MOSFETs, which typically require a drift layer thickness exceeding 10 μm. The results mark a major step of cost-effective GaN-on-Si vertical MOSFETs towards 1200 V-class power applications.

IV. CONCLUSION

In this letter, we demonstrate 1200 V fully-vertical GaN-on-Si trench MOSFETs with fluorine ion-implanted termination (FIT) technique. The FIT structure replaces the normally used mesa etching termination and effectively isolates the discrete FIT-MOS, which eliminates the electric field crowding effect in the MET-MOS, therefore resulting in state-of-the-art breakdown voltage of 1277 V and $BFOM$ of 291 MW/cm². The fluorine implantation termination technique holds great promise for the applications of vertical GaN trench MOSFETs in kV-class power systems.

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